

# AutoBench Service — Developer Guide

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Hand-maintained, unlike the generated `results/12run-*.md` files which stamp themselves. Bump the line above when you edit this guide.

A task-oriented guide to driving the AutoBench Service over its RESTful API. Every `curl` below is derived from the flows we exercised end-to-end (gsm8k single-turn, tau2 multi-turn) on the `ykt3` and `kind-rossoctl` clusters.

- Reference the machine-readable contract in `openapi.json` / `openapi.yaml`.
- Design rationale (what the Service can and cannot enact) lives in `SERVICE_DESIGN_DECISIONS.md` and `KUBECTL_DEPENDENCY_INVENTORY.md`.

**In a hurry?** §2 gets you a token, §6.0 is one benchmark run start to finish as copy-pasteable `curl`, and §6.1 is the same thing as a single command.

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## 1. Mental model (read this first)

The Service is **pure-Python and HTTP-only**. It never calls a cluster API (no `kubectl`, no `kubeconfig`). It talks to the Rossoctl/kagenti backend over HTTPS and to Keycloak for tokens. Two consequences shape everything below:

### 1. Two independent tokens.

- The **caller JWT** you present in `Authorization: Bearer ...` is used *only* for attribution ( `preferred_username` ) and routing ( `iss` selects which instance config applies). It is **never forwarded** upstream.
- For each request the Service mints its **own** ROPC token (the instance's `benchmarker` credential, password grant) and presents *that* to Rossoctl.

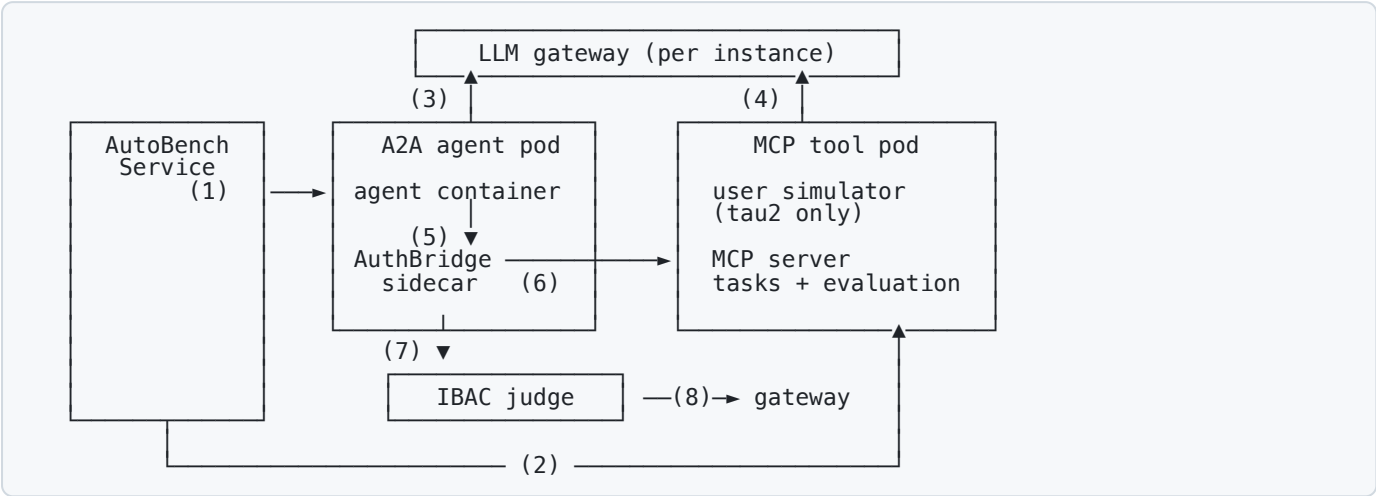
2. **The Service enacts only what HTTP allows.** It deploys workloads (agents/tools), runs benchmarks, reads reports, and exports to S3. It **cannot** create cluster Secrets or overlay per-agent ConfigMaps — those are provisioned out-of-band, and the Service reports on them (see §3 onboarding and the `424` / `422` responses).

## Endpoint map

| Group            | Endpoints                                                                                                                                | Auth                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Identity         | GET <code>/hello</code> , GET <code>/healthz</code> (public, unschematized)                                                              | caller JWT              |
| Config           | GET <code>/config</code> , PUT <code>/config</code>                                                                                      | <b>benchmarker only</b> |
| Discovery        | GET <code>/namespaces</code> , GET <code>/benchmarks</code> , GET <code>/benchmarks/{name}</code>                                        | caller JWT              |
| Raw workloads    | GET/POST <code>/agents</code> , GET/DELETE <code>/agents/{ns}/{name}</code> , same for <code>/tools</code>                               | caller JWT              |
| Benchmark deploy | POST <code>/benchmarks/{name}/deploy</code> , DELETE <code>/benchmarks/{name}/deploy</code> , GET <code>/benchmarks/{name}/status</code> | caller JWT              |
| Run lifecycle    | POST <code>/benchmarks/{name}/runs</code> , GET <code>.../runs</code> , GET <code>.../runs/{run_id}</code>                               | caller JWT              |
| Reporting        | GET <code>.../runs/{run_id}/report</code> , GET <code>/benchmarks/{name}/report</code>                                                   | caller JWT              |

The run-time data path

The endpoint map above is the **control** plane — what you call. During a run a different set of legs carries the actual work, and none of them are HTTP calls you make. Chart 6 of [AutoBench.pptx](#) draws this; the same eight legs in text:



| # | leg                                                                                | is the sidecar on it?                                                        |
|---|------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 1 | Service → agent container: send_prompt, <b>once per task</b>                       | <b>yes, inbound</b> — logged as a2a-parser; isAction=false, so no judge call |
| 2 | Service → MCP server: list_tasks, create_session, evaluate_session, delete_session | no — a different pod, and the Service does not dial through the proxy        |
| 3 | agent → LLM gateway: N chat calls per task                                         | <b>no</b> — see below                                                        |
| 4 | MCP tool → LLM gateway: tau2's user simulator (a <i>second</i> model)              | no — a different pod                                                         |
| 5 | agent → sidecar: the agent's own MCP tool calls                                    | <b>this is the interception point</b>                                        |
| 6 | sidecar → MCP server: once the call is authorized                                  | outbound mcp-parser, method=tools/call isAction=true                         |
| 7 | sidecar → IBAC judge: one call per isAction tool call                              | —                                                                            |
| 8 | judge → LLM gateway: the verdict is itself an LLM call                             | —                                                                            |

Three things about this shape are not guessable from the API surface:

**The agent has its own MCP connection, separate from the Service's.** `MCP_URL` is injected into the agent's env at deploy time (always `svc.cluster.local`, since agent→tool is intra-cluster even on a cross-cluster run). The Service opens and grades the session (leg 2); the agent's tool calls (legs 5–6) mutate that same session's state, correlated by `session_id` — which travels as A2A request metadata, *not* in the prompt text. So grading reads tool-side state, and the runner discards whatever the agent replies: an agent that answers in prose without making the submitting tool call fails the task.

**Interception is a blanket forward proxy plus an allowlist, not a tool-aware hook.** With AuthBridge enabled the operator injects `HTTP_PROXY=HTTPS_PROXY=http://127.0.0.1:8081` — the sidecar — into the agent pod, which would capture *every* outbound HTTP call it makes, inference included. What keeps leg 3 out of it is the instance config's `no_proxy`, which names the LLM gateway host (and the OTEL collector, and

Keycloak). IBAC additionally sets `judge_inference: false`, so even proxied inference would not be judged. Both would have to change for the agent's model calls to be authorized.

**A ready sidecar proves nothing about enforcement.** Because the switch is a proxy-env allowlist, a plausible-looking config can leave the outbound plugins seeing no traffic at all: setting `workload_llm.disable_proxy: true` makes the Service inject `HTTP_PROXY=""` first, and the operator only adds a var when it is *absent*, so it skips `HTTP_PROXY` while still setting `HTTPS_PROXY`. `MCP_URL` is `http://`, so tool calls went direct — and a `no_proxy` containing `.svc.cluster.local` excluded them a second time. The sidecar was ready, the pipeline rendered, the judge was called **zero** times. Keep `disable_proxy` off and keep the *wildcards* out of `no_proxy` while keeping the named hosts in it, then verify by **counting judge completions across a run** — IBAC emits no log line of its own, so sidecar logs cannot tell you. `initialize` and `tools/list` are `isAction=false` and legitimately never reach the judge, so a zero count only means something if real tool calls occurred. Cost figures for each layer: [PLUGIN\\_OVERHEAD.md](#).

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## 2. Getting a caller token

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The caller JWT is a normal Keycloak token from the instance's realm. Obtain it via the password grant (the same Direct-Access-Grants flow the dev/test users use). The realm and token endpoint are derived from the instance `iss` (`<iss-origin>/realms/<realm>/protocol/openid-connect/token`).

```
# --- Environment (pick the block for your target cluster) ---
# Option A – ykt3 (OpenShift; realm/client "kagenti"), the cluster the e2e runs were validated
# on:
export SVC="https://autobench.apps.ykt3.example.com"      # AutoBench Service base URL
export KC="https://keycloak.apps.ykt3.example.com"        # Keycloak base (iss origin)
export REALM="kagenti"                                   # realm from the iss
export KC_CLIENT="kagenti"                               # public client with Direct Access
# Grants
# Option B – kind-rossoctl (local; realm/client "rossoctl"):
# The instance's iss is http://keycloak.localtest.me:8080/realms/rossoctl, but in-cluster the
# Service reaches Keycloak via keycloak_backchannel_url. You still fetch YOUR token from the
# externally reachable route below (localtest.me resolves to 127.0.0.1).
# export SVC="http://autobench.localtest.me:8080"
# export KC="http://keycloak.localtest.me:8080"
# export REALM="rossoctl"
# export KC_CLIENT="rossoctl"
# Attribution matters: config endpoints require preferred_username == "benchmarker".
# Any valid realm user works for deploy/run/report.
# Never hardcode the password – supply it out-of-band (env, secret manager, or an
# interactive prompt). e.g.: read -rs -p "benchmarker password: " KC_PASS; export KC_PASS
export KC_USER="benchmarker"
export KC_PASS="${KC_PASS:?set KC_PASS in your environment; do not commit it}"
export TOKEN=$(curl -s "$KC/realms/$REALM/protocol/openid-connect/token" \
  -d grant_type=password \
  -d client_id="$KC_CLIENT" \
  -d username="$KC_USER" \
  -d password="$KC_PASS" \
  | python3 -c 'import sys,json; print(json.load(sys.stdin)["access_token"])')
# Sanity check: the Service echoes the validated claims + which instance you routed to.
curl -s "$SVC/hello" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```

GET /hello returns `{iss, preferred_username, rossoctl_base_url, claims}`. If you get `401`, the token is missing/invalid; `403 issuer not in scope` means no instance file matches the token's `iss` (see §3.1).

**Why the kind block differs:** in-cluster the public `iss` host is unreachable, so the instance file sets `keycloak_backchannel_url` and the Service dials *that* for JWKS/ROPC. This only affects the Service's own dialing — your caller token still comes from the externally reachable Keycloak route (Option B above).

### 3. Onboarding a benchmark (one-time, per cluster/instance)

These steps are **out-of-band** — the Service verifies them but cannot perform them.

#### 3.1 Instance config file ( `instances/<encoded-iss-host>.json` )

One file per instance keys the whole thing to the `iss`. Loaded at startup from `settings.instances_dir`. Shape:

```
{
  "iss": "https://keycloak.apps.ykt3.example.com/realms/kagenti",
  "keycloak_backchannel_url": null,
  "rossoctl_base_url": "https://kagenti-backend.kagenti-system.svc.cluster.local:8000",
  "service_credential": {
    "client_id": "kagenti",
    "client_secret": "",
    "username": "benchmarker",
    "password": "<benchmarker-password>"
  },
  "mlflow": { "tracking_url": null },
  "s3": { "bucket": null },
  "mcp_endpoint_template": null,
  "agent_endpoint_template": null,
  "workload_otel": null
}
```

- `service_credential` is the ROPC identity the Service presents to Rossoctl (must exist in the realm with `firstName` / `lastName` set, or Keycloak 400s with "Account is not fully set up").
- `mcp_endpoint_template` / `agent_endpoint_template` are only needed when the workloads live on a **different** cluster reachable via external routes (use `{service}` / `{namespace}` placeholders). Leave `null` for co-located in-cluster workloads.
- `mlflow` / `s3` can be seeded here or set later via `PUT /config` (§4).

#### 3.2 Infrastructure resources (baked into the benchmark definitions)

The Service sends CPU/memory requests+limits in the create body, so no post-create patch is needed. Current values (from `benchmarks/registry.py`):

| Workload  | requests         | limits      |
|-----------|------------------|-------------|
| MCP tool  | 500m CPU / 512Mi | 4 CPU / 4Gi |
| A2A agent | 500m CPU / 512Mi | 4 CPU / 2Gi |

### 3.3 Workload secrets (provisioned out-of-band as cluster Secrets)

The Service has **no Secrets API**. It references these by name in the deploy body; if a Secret is missing the workload never becomes Ready, and the run precheck returns **424** naming exactly what to provision.

| Benchmark               | Secret (name → key)    | Consumed by                  | Purpose                    |
|-------------------------|------------------------|------------------------------|----------------------------|
| gsm8k                   | hf-secret → hf-token   | tool                         | HuggingFace dataset access |
| gsm8k / tau2 / appworld | openai-secret → apikey | agent (+ tau2/appworld tool) | LiteLLM API key            |

Provision them on the workload cluster before deploying, e.g.:

```
kubectl -n team1 create secret generic hf-secret --from-literal=hf-token="$HF_TOKEN"
kubectl -n team1 create secret generic openai-secret --from-literal=apikey="$LITELLM_KEY"
```

**openai-secret** must exist before you run the Service — provision it out-of-band. On clusters where the rossoctl platform chart owns the agent namespaces (e.g. `team1`), the chart only manages `openai-secret` when its Helm value `secrets.openaiApiKey` is set. Leave that value **empty** and create the Secret yourself (command above / a secret manager / External Secrets), so the credential lives outside Helm values and a later `helm upgrade` **cannot** overwrite it with an empty key. When the value is empty the chart skips the Secret entirely and its install NOTES print a `WARNING: secrets.openaiApiKey is NOT set` reminder. If you instead let the chart template the key, every `helm upgrade` re-applies whatever is in the release values — an empty value there silently zeroes the key mid-flight and every LLM call fails until the Secret is restored and the workload pods are `rollout restart`ed (pods read `apikey` via `secretKeyRef` only at startup).

### 3.4 MLflow + OTEL collector (optional, for reports)

Reporting is fail-soft: if MLflow client-creds aren't configured, runs still succeed and export to S3, but `report.ndjson` is empty and the report endpoints return **409**. To enable, provision an OTEL collector (forwards agent spans to MLflow) out-of-band and set the MLflow read creds via `PUT /config`.

## 4. Instance-specific Service config ( /config )

`GET /config` are **benchmarker-only** (`preferred_username == "benchmarker"`; else **403**). They set only what the Service itself enacts — MLflow (read side) and S3. Attempting to set a workload credential is rejected with **422** (`extra="forbid"`). Secrets are redacted in responses.

```
# View effective config for your instance (file defaults + any overrides), secrets redacted.
curl -s "$SVC/config" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
# Point the Service's result sink at an S3 bucket + wire up MLflow read creds.
curl -s -X PUT "$SVC/config" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{
    "s3": {
      "bucket": "rossoctl-benchmarking",
      "region": "us-east-1",
      "access_key_id": "AKIA...REDACTED",
      "secret_access_key": "REDACTED",
      "public_read": false
    }
  }'
```

```

    },
    "mlflow": {
      "tracking_url": "https://mlflow.apps.ykt3.example.com",
      "client_id": "mlflow-client",
      "client_secret": "REDACTED",
      "token_url": "https://keycloak.apps.ykt3.example.com/realms/kagenti/protocol/openid-connect/token",
      "insecure_tls": false
    }
  }
}'

```

Rejected example (workload cred → 422 ):

```

curl -s -o /dev/null -w '%{http_code}\n' -X PUT "$SVC/config" \
-H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
-d '{"hf-secret": "..."}' # -> 422

```

## 5. Benchmark lifecycle

The happy path is: **deploy** → **wait for Ready** → **run** → **poll** → **report** → **download from S3**.

Every block below is copy-pasteable once these four variables are exported (see §2 for how to get the token; §6.1 wraps this whole section in a single command if you would rather not paste):

```

export SVC="https://autobench-rossoctl-system.apps.ykt3.hcp.res.ibm.com" # Service base URL
export BENCH=gsm8k # gsm8k | tau2 | appworld
export SCOPE="namespace=team1&agent=tool_calling&experiment=default"
export TOKEN="" # from §2; never echo this
# Self-signed OpenShift route? add -k to every curl, or: export CURL_OPTS=-k

```

### 5.0 Discover what's available

```

curl -s "$SVC/benchmarks" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
curl -s "$SVC/benchmarks/gsm8k" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
curl -s "$SVC/namespaces" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool

```

Three benchmarks are registered: `gsm8k` (single-turn), `tau2` (multi-turn, runs a server-side user-simulator LLM), `appworld` (registered; see §7).

### 5.1 Deploy (create the MCP tool + A2A agent)

`POST /benchmarks/{name}/deploy` creates both the shared MCP tool (`exgentic-mcp-<name>`) and the agent (`exgentic-a2a-<agent>-<name>[-<experiment>]`). Returns `201`.

```

curl -s -X POST "$SVC/benchmarks/gsm8k/deploy" \
-H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
-d '{
  "agent": "tool_calling",
  "namespace": "team1",
  "experiment": "default"
}' | python3 -m json.tool

```



Request fields ( `DeployBenchmarkRequest` ):

| Field              | Default      | Meaning                                                                                                                                                                             |
|--------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| agent              | tool_calling | agent flavor (only tool_calling today)                                                                                                                                              |
| model              | null         | <b>deploy-time</b> model, baked into agent env; null → benchmark default (openai/Qwen3.6-35B-A3B)                                                                                   |
| namespace          | team1        | target namespace                                                                                                                                                                    |
| experiment         | default      | non-default suffixes the agent name so variants coexist                                                                                                                             |
| authbridge_enabled | false        | inject the AuthBridge sidecar with the <b>cluster-default</b> pipeline (layer-2 knob)                                                                                               |
| plugin_preset      | null         | layer-3 preset (auth-only ibac-only full); forwarded to the backend as pluginPreset → AgentRuntime.spec → operator renders the per-agent pipeline. Requires authbridge_enabled=true |
| plugins            | null         | per-plugin policy overrides as ["NAME:POLICY"] tokens (POLICY = enforce observe off); forwarded as plugins. Requires authbridge_enabled=true                                        |
| on_error           | null         | chain-default policy (enforce observe off); forwarded as onError. Requires authbridge_enabled=true                                                                                  |
| plugin_config_file | null         | <b>rejected with 422</b> — a local filesystem path with no HTTP analog; use plugin_preset/plugins/on_error instead                                                                  |

**Model swap** (run #4 pattern): the deploy endpoint always (re)creates the *shared* tool, so re-deploying with a new model 409s on the existing tool. Deploy the model-swapped agent alone via `POST /agents` under a distinct experiment instead:

```
# Generate the agent body with build_agent_request semantics, then POST /agents directly.
curl -s -X POST "$SVC/agents" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{
    "name": "exgentic-a2a-tool-calling-gsm8k-azmini",
    "namespace": "team1",
    "container_image": "ghcr.io/exgentic/exgentic-a2a-tool_calling",
    "image_tag": "latest",
    "env_vars": [
      {"name": "MCP_URL", "value": "http://exgentic-mcp-gsm8k-mcp.team1.svc.cluster.local:8000/mcp"},
      {"name": "LLM_MODEL", "value": "openai/Azure/gpt-4o-mini"},
      {"name": "EXGENTIC_SET_AGENT_MODEL", "value": "openai/Azure/gpt-4o-mini"},
      {"name": "OPENAI_API_KEY", "value_from": {"secret_key_ref": {"name": "openai-secret",
"key": "apikey"}}}
    ],
    "service_ports": [{"name": "http", "port": 8080, "target_port": 8000}]
  }' | python3 -m json.tool
```

## 5.2 Wait until Ready

```
curl -s "$SVC/benchmarks/gsm8k/status?namespace=team1&agent=tool_calling&experiment=default" \
  -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```



Returns `tool_ready` / `agent_ready` booleans plus raw `readyStatus` . Poll until both are `true` . If a workload stays not-Ready, the run precheck (§5.3) will name the missing Secret.

### 5.3 Submit a run

`POST /benchmarks/{name}/runs` runs a **cluster-API-free precheck** (tool+agent deployed and Ready) then returns `202` with a `run_id` . Prechecks:

- `409` — not deployed ( `POST .../deploy` first).
- `424` — deployed but not Ready; the message names required Secret(s) (e.g. `hf-secret` ).

```
export RUN=$(curl -s -X POST "$SVC/benchmarks/gsm8k/runs" \
-H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
-d '{
  "agent": "tool_calling",
  "namespace": "team1",
  "experiment": "default",
  "max_tasks": 10,
  "max_parallel_sessions": 1,
  "timeout_seconds": 600
}' | python3 -c 'import sys,json; print(json.load(sys.stdin)["run_id"])')
echo "run_id=$RUN"
```

Run fields ( `RunRequest` ) are all **run-time** knobs:

| Field                                       | Default | Notes                                                                                       |
|---------------------------------------------|---------|---------------------------------------------------------------------------------------------|
| <code>max_tasks</code>                      | 1       | number of benchmark tasks to evaluate; <b>capped by the task pool, silently</b> — see below |
| <code>max_parallel_sessions</code>          | 1       | concurrency                                                                                 |
| <code>timeout_seconds</code>                | 300     | whole-run wall-clock ceiling; raise for large tau2 runs                                     |
| <code>agent / namespace / experiment</code> | —       | must match a deployed agent                                                                 |
| <code>model</code>                          | null    | <b>not</b> forwarded per-session; model is fixed at deploy time                             |

**What `max_tasks: N` actually means.** `N independent episodes against one deployment` — not one long session, and not `N` deploys. The runner fetches the task list, slices off the first `N`, and for each id runs the same three-step sequence — `MCP.CreateSession` → `Agent.Call` (A2A `send_prompt`) → `Evaluator.Evaluate` — inside its own `Agent.Session` OTEL span keyed by that `task_id` ( `runner/engine.py` ). Nothing carries over between tasks: no shared conversation, no shared MCP session, no ordering dependence. Failures are isolated per task (each `_one` has its own `asyncio.timeout` and the batch is gathered with `return_exceptions=True` ), so one wedged task costs you that task, not the run — and the summary is republished after **every** task, so even a run the wall clock kills leaves usable partial results. That independence is what makes `pass_rate = evaluated_pass / total` a rate rather than an average of correlated trials.

`max_tasks` is **how many**; `max_parallel_sessions` is **how many at once** (a separate semaphore). They are orthogonal: `max_tasks: 50, max_parallel_sessions: 4` is fifty problems with four in flight. The agent and MCP pods are shared by all `N` tasks, which is what makes a multi-task run cheap relative to its

task count — and also why per-task *latency* from a short run is dominated by the first concurrency wave's warm-up.

**max\_tasks: 1 against a 114-task pool is not an error either** — it is the same slice seen from the other side. You get `task_id 0` and only `task_id 0`; the other 113 are never created, and nothing reports them as skipped. Both directions of mismatch are silent, and `summary.total` is the one field that tells you what actually ran: your number when you under-ask, the pool size when you over-ask. The habit to build: **a 1-task run is a smoke test, not a measurement**. It proves the deploy, auth, LLM reachability and telemetry path work end to end — that is exactly why leg #1 of the canonical matrix is one gsm8k task — but it says nothing about capability, because it is a single pass/fail on the *same* problem every time. Because the slice is a prefix, every leg of a benchmark re-runs that first task: legs #1  $\subset$  #2  $\subset$  #3, and #5–#8 are all the same first five. That is deliberate — it is what makes legs comparable across configurations and clusters — but it has a sharp edge: legs sharing prompts also share the LLM gateway's completion cache, so they must be spaced (§6.4).

**max\_tasks above the benchmark's task pool is not an error**. The runner asks the MCP for the task list and slices it — `task_ids[:max_tasks]` — so a request for more tasks than exist yields the whole pool with no warning, and `summary.total` then reports the pool size rather than what you asked for. There is no upper bound on the field to catch it. The pools differ by an order of magnitude (gsm8k ~8.5K from HuggingFace, tau2 114 in the default `retail` domain, appworld grouped scenarios), so check the size before requesting a large run: [BENCHMARKS\\_PRIMER.md](#) has them in one table. Selection is **deterministic** — the first `max_tasks` of the pool — which is what makes a smaller run's tasks a prefix of a larger one's.

**Timeout tuning (learned e2e)**: tau2 with `max_tasks=20 max_parallel_sessions=4` exceeded the 900s default and failed; re-running with `timeout_seconds=1800` succeeded (~915s).

## 5.4 Poll run status / list runs

```
# One run's full state (status, summary, per-task results, artifacts once exported).
curl -s "$SVC/benchmarks/gsm8k/runs/$RUN" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
# All runs for this benchmark scoped to your instance.
curl -s "$SVC/benchmarks/gsm8k/runs" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```

`status` moves `pending` → `running` → `succeeded|failed`. On success, `summary` carries `{total, succeeded, evaluated_pass, pass_rate, wall_seconds}`. A simple poll loop:

```
# Terminal statuses are succeeded | failed | error | cancelled – a loop that waits only for
# succeeded/failed spins forever on the other two.
while ;; do
  S=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer $TOKEN" \
    | python3 -c 'import sys,json; print(json.load(sys.stdin)["status"])')
  echo "status=$S"
  case "$S" in succeeded|failed|error|cancelled) break ;; esac
  sleep 10
done
# The one-line verdict.
curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer $TOKEN" >
  /tmp/run.json
python3 - <<'EOF'
import json
```

```
d = json.load(open("/tmp/run.json")); s = d["summary"]
print("%s pass_rate=%s %s/%s wall=%.0fs" % (
    d["status"], s["pass_rate"], s["evaluated_pass"], s["total"], s["wall_seconds"]))
EOF
```

## 5.5 Get results (report)

Two report views, both reading structured records from MLflow (return 409 if MLflow isn't configured for the instance — see §3.4):

```
# Per-run report: records filtered to this run's session ids, plus its S3 artifacts.
curl -s "$SVC/benchmarks/gsm8k/runs/$RUN/report" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
# Experiment rollout: time-windowed aggregates across an experiment's runs.
curl -s "$SVC/benchmarks/gsm8k/report?experiment=default&window_h=3" \
-H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```

Per-run report is {run\_id, benchmark, experiment, trace\_count, records[], artifacts[]}; each record (MLflowTraceRecord) has per-session timing breakdown, LLM/tool latencies + token counts, infra CPU/mem, and the evaluation outcome.

## 5.6 Download output files from S3

When the instance has an S3 bucket set, each completed run is exported (fail-soft) and its objects appear in the run state under artifacts[] (and in the per-run report). Layout:

```
<prefix>/<preferred_username>/<encoded-iss>/<benchmark>/<run_id>/
run.json           # the run summary (always written)
report.ndjson      # one record per task (empty if MLflow unconfigured)
report.parquet     # analytics-friendly (only when there are records)
token_report.ndjson # lean per-task token view, projected from the records
token_report.parquet
span_report.ndjson  # one row per OTEL span per task, incl. the span name
span_report.parquet
manifest.json       # self-describing index of the above (never lists itself)
```

span\_report.\* is the evidence layer under the other two: report / token\_report give per-task counts, span\_report gives the spans those counts were derived from, so "did this task really make one model call, or did we lose a span?" is answerable from the artifacts alone. Its columns are a fixed whitelist (mlflow\_report.SPAN\_ROW\_KEYS) — span attributes can carry prompts, and these objects are public-read, so nothing outside that list is published.

### The S3 URL is fully determined — you can construct it

Objects are addressed as <url-root>/<key>, and the key is the layout above. For the bucket these runs use:

|          |                                                                                     |
|----------|-------------------------------------------------------------------------------------|
| URL root | https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/                           |
| Key      | <s3.prefix>/<preferred_username>/<encoded-iss-host>/<benchmark>/<run_id>/<artifact> |

So a real, working URL — anonymously readable, no credentials, no AWS CLI:

```
https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/ykt3-to-ykt2/benchmark/keycloak-  
keycloak.apps.ykt2.hcp.res.ibm.com-realms-rossoctl/gsm8k/20260902215628-  
79e0713a/report.ndjson
```

`s3.prefix` is per instance (`ykt3-to-ykt2/` on the OpenShift instance, `kind/` on KinD), and the issuer host is scheme-stripped with `./ : / /` replaced by `-`. **The bucket is public and anonymously listable**, so treat everything exported as published.

```
# The authoritative list - name, format, size and the full URL, straight from the run state.  
curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer $TOKEN" >  
  /tmp/run.json  
python3 - <<'EOF'  
import json  
for a in json.load(open("/tmp/run.json"))["artifacts"]:  
    print("%-8s %9d %s" % (a["format"], a["size_bytes"], a["url"]))  
EOF  
# Download every artifact by URL - plain curl, no AWS credentials needed.  
PREFIX=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer $TOKEN"  
  \  
  | python3 -c 'import sys,json; print(json.load(sys.stdin)["artifacts_prefix"])')  
mkdir -p "/tmp/autobench/$PREFIX" && cd "/tmp/autobench/$PREFIX"  
for f in run.json report.ndjson report.parquet token_report.ndjson token_report.parquet \  
  span_report.ndjson span_report.parquet manifest.json; do  
    curl -sS -O "https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/$PREFIX/$f"  
done && ls -la  
# Or discover the set from the manifest, which indexes every object but itself.  
curl -s "https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/$PREFIX/manifest.json" \  
  | python3 -m json.tool  
# Private bucket instead of public_read? Then use the AWS CLI with credentials.  
aws s3 cp "s3://rossoctl-benchmarking/$PREFIX/" ./ --recursive
```

## 5.7 Tear down

```
curl -s -X DELETE "$SVC/benchmarks/gsm8k/deploy?  
  namespace=team1&agent=tool_calling&experiment=default" \  
  -H "Authorization: Bearer $TOKEN" -o /dev/null -w '%{http_code}\n' # -> 204
```

Deletes the agent and the shared tool. For a model-swap variant deployed via `POST /agents`, delete it with `DELETE /agents/{namespace}/{name}` (the shared tool is left for other experiments).

## 6. End-to-end examples (validated flows)

### What you need on the client side

**Python version.** The client side is deliberately undemanding: §6.0 needs only `curl` plus any `python3` for JSON formatting, and `autobench-cli` (§6.1) imports nothing beyond the standard library. Versions we have actually run:

| Python  | Where it was used                                                   | Result |
|---------|---------------------------------------------------------------------|--------|
| 3.12.12 | recommended for a client venv — matches the workload images exactly | ✓      |

| Python  | Where it was used                                                    | Result |
|---------|----------------------------------------------------------------------|--------|
| 3.12.14 | the Service container itself (FROM python:3.12-slim)                 | ✓      |
| 3.11.14 | the repo's own venv; the declared floor (requires-python = ">=3.11") | ✓      |
| 3.14.3  | the laptop that drove both 12-run matrices                           | ✓      |

Anything  $\geq 3.11$  is fine. 3.12.12 is the tidiest choice because it is exactly what the exgentic agent and MCP images run, so your driver matches the workload interpreter — but nothing in the client depends on it, and the `curl` path in §6.0 is version-insensitive.

**Do you have to clone the repo?** For §6.0, **no** — it is `curl` and `python3` only, so a token and a Service URL are all you need. For §6.1 you need the package that provides the `autobench-cli` entry point, but a manual clone is still optional: the repository is public, so install it straight from git.

```
# No clone. Client only: --no-deps, because the CLI is stdlib-only and pulls in nothing else.
uv venv --python 3.12.12 && source .venv/bin/activate
uv pip install "rossoctl-autobench @ git+https://github.com/rossoctl/autobench" --no-deps
# Already have a clone? Point at the repo, NOT at `.` — `-e .` looks for pyproject.toml in the
# CURRENT directory, so running it from a scratch/testbed folder fails with
# "does not appear to be a Python project".
uv pip install -e /path/to/autobench --no-deps           # -e: git pull updates the CLI in place
```

Drop `--no-deps` only if you also want to run the Service or its test suite from that venv; it adds fastapi, boto3, pydantic and the MCP/A2A SDKs, none of which the client needs. Keep the venv outside a cloud-synced folder (Box/Dropbox/iCloud) — a venv is thousands of small files, and sync tools are slow and occasionally destructive with it.

Where the source ends up

`<venv>/bin/autobench-cli` is **not** the source — it is a generated shim that does `from autobench.cli import main`. Where the real `cli.py` lives depends on how you installed, which also decides whether a `git pull` reaches you:

| Install                                                                                      | cli.py lives in                                                                                          | Picks up upstream changes?                        |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <code>uv pip install "...@ git+https://..."</code>                                           | <code>&lt;venv&gt;/lib/python3.X/site-packages/autobench/</code> (a <b>copy</b> , built from one commit) | No — pinned. Re-run with <code>--reinstall</code> |
| <code>uv pip install -e /path/to/autobench</code>                                            | <code>/path/to/autobench/src/autobench/</code> (the clone; only a finder hook is installed)              | Yes, immediately                                  |
| <code>uv tool install --from /path/to/autobench autobench-service</code>                     | <code>\$(uv tool dir)/rossoctl-autobench/lib/python3.X/site-packages/autobench/</code>                   | No — reinstall the tool                           |
| no install, <code>PYTHONPATH=&lt;repo&gt;/src</code><br><code>python -m autobench.cli</code> | the clone itself                                                                                         | Yes, immediately                                  |

Never guess the path — ask the interpreter that is actually running it:

```
python -c "import autobench.cli as m; print(m.__file__)"
```

Note that every option installs the **whole distribution**, server modules included ( `app.py` , `routes/` , `runner/` , `s3_export.py` ), because they ship together. With `--no-deps` the client half still works while `import autobench.app` does not — the third-party dependencies were skipped, not the files. That the client keeps working is the point, and you can confirm it:

```
python -c "
import autobench.cli, sys
print([m for m in ('fastapi','httpx','boto3','pydantic') if m in sys.modules] or 'no server deps
loaded')"
```

## 6.0 Run #1 start to finish, on the ykt3 Service driving ykt2 workloads

Run #1 of the canonical matrix: gsm8k, 1 task, no gateway, no plugins. Every command below was executed exactly as written; the `run_id` and numbers are from that run. Paste the block, then the steps.

```
# ---- 0. environment (the cross-cluster instance: Service on ykt3, workloads on ykt2/team1) ----
-
export SVC="https://autobench-rossoctl-system.apps.ykt3.hcp.res.ibm.com"
export KC="https://keycloak-keycloak.apps.ykt2.hcp.res.ibm.com" # the instance's iss origin
export REALM=rossoctl KC_CLIENT=rossoctl KC_USER=benchmarker
export BENCH=gsm8k
export SCOPE="namespace=team1&agent=tool_calling&experiment=default"
export CURL_OPTS=-k # ykt3's route serves a self-signed cert
read -rs -p "benchmarker password: " KC_PASS; echo # never put this in a file you commit
# ---- 1. caller token ----
export TOKEN=$(curl -s $CURL_OPTS "$KC/realms/$REALM/protocol/openid-connect/token" \
-d grant_type=password -d client_id="$KC_CLIENT" \
-d username="$KC_USER" -d password="$KC_PASS" \
| python3 -c 'import sys,json; print(json.load(sys.stdin)["access_token"])')
unset KC_PASS
curl -s $CURL_OPTS "$SVC/hello" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
# -> {"iss": ".../realms/rossoctl", "preferred_username": "benchmarker", ...}
# ---- 2. deploy the MCP tool + A2A agent (idempotent: delete first) ----
curl -s $CURL_OPTS -X DELETE "$SVC/benchmarks/$BENCH/deploy?$SCOPE" \
-H "Authorization: Bearer $TOKEN" -o /dev/null -w 'delete -> %{http_code}\n' # 204 or 404
sleep 10
curl -s $CURL_OPTS -X POST "$SVC/benchmarks/$BENCH/deploy" \
-H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
-d '{"agent":"tool_calling","namespace":"team1","experiment":"default"}' \
-o /dev/null -w 'deploy -> %{http_code}\n' # 201
# ---- 3. wait until BOTH are Ready, four polls in a row ----
# One reading is not enough: an agent whose MCP is not serving yet exits and
# CrashLoopBackOffs, so readiness flaps. A run accepted mid-flap records 0 tasks.
STABLE=0; until [ "$STABLE" -ge 4 ]; do
  R=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/status?$SCOPE" -H "Authorization: Bearer
    $TOKEN" \
    | python3 -c 'import sys,json; d=json.load(sys.stdin); print(int(bool(d["tool_ready"] and
    d["agent_ready"])))')
  [ "$R" = 1 ] && STABLE=$((STABLE+1)) || STABLE=0
  echo "ready=$R stable=$STABLE"; sleep 10
done
# On OpenShift the Route 502s for ~10s AFTER the Service reports Ready, so gate on the card too:
AGENT=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/status?$SCOPE" -H "Authorization: Bearer
  $TOKEN" \
  | python3 -c 'import sys,json; print(json.load(sys.stdin)["agent_name"])')
until curl -sf -o /dev/null "https://$AGENT-team1.apps.ykt2.hcp.res.ibm.com/.well-known/agent-
card.json"; do
```



```

    echo "waiting for the agent card"; sleep 5
done; sleep 15
# ---- 4. submit the run ----
export RUN=$(curl -s $CURL_OPTS -X POST "$SVC/benchmarks/$BENCH/runs" \
-H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
-d '{"agent":"tool_calling","namespace":"team1","experiment":"default",
    "max_tasks":1,"max_parallel_sessions":1,"timeout_seconds":120}' \
| python3 -c 'import sys,json; print(json.load(sys.stdin)["run_id"])')
echo "run_id=$RUN" # e.g. 20260902215628-79e0713a
# ---- 5. poll to a terminal status, then print the verdict ----
while ;; do
    S=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer $TOKEN" \
| python3 -c 'import sys,json; print(json.load(sys.stdin)["status"])')
    echo "status=$S"; case "$S" in succeeded|failed|error|cancelled) break ;; esac; sleep 10
done
# -> succeeded, summary
    {"total":1,"succeeded":1,"evaluated_pass":1,"pass_rate":1.0,"wall_seconds":5.4}
# Note ~100 spans for ONE gsm8k task: ~91 are A2A/HTTP framework internals.
# ---- 6. artifacts: list the S3 URLs, then mirror them locally ----
export PREFIX=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer
$TOKEN" \
| python3 -c 'import sys,json; print(json.load(sys.stdin)["artifacts_prefix"])')
echo "$PREFIX"
# -> ykt3-to-ykt2/benchmark/keycloak-keycloak.apps.ykt2.hcp.res.ibm.com-realms-
rossocctl/gsm8k/20260902215628-79e0713a
mkdir -p "/tmp/autobench/$PREFIX" && (cd "/tmp/autobench/$PREFIX" && \
for f in run.json report.ndjson report.parquet token_report.ndjson token_report.parquet \
span_report.ndjson span_report.parquet manifest.json; do
    curl -sS -O "https://rossocctl-benchmarking.s3.us-east-1.amazonaws.com/$PREFIX/$f"; done &&
ls -la)
# ---- 7. analyse: per-task tokens, and whether the telemetry is trustworthy ----
cd "/tmp/autobench/$PREFIX"
python3 - <<'EOF'
import json
rows = [json.loads(l) for l in open("report.ndjson") if l.strip()]
print("%-6s %-38s %7s %7s %5s %6s %s" % ("task", "model", "in", "out", "llm", "tool",
    "passed"))
for r in sorted(rows, key=lambda x: str(x["task_id"])):
    print("%-6s %-38s %7d %7d %5s %6s %s" % (
        r["task_id"], r["model"], r["llm_input_tokens"], r["llm_output_tokens"],
        r["llm_count"], r["tool_count"], r["evaluation_result"]))
# llm<=1 alongside tool>=2 is impossible -- every tool call needs a preceding model turn -- so
it
# means the usage-bearing span was lost and the token counts understate the real cost.
bad = [r for r in rows if (r["llm_count"] or 0) <= 1 and (r["tool_count"] or 0) >= 2]
print("lost token attribution:", len(bad), "of", len(rows))
EOF
# -> 0      openai/Azure/gpt-5-mini-2025-08-07      320      471      2      1      True
# lost token attribution: 0 of 1
# Which spans did the task actually invoke? A healthy task shows TWO chat spans -- the
# max_tokens=1 capability probe plus the real call. One means the real call's span was lost.
python3 - <<'EOF'
import collections, json
rows = [json.loads(l) for l in open("span_report.ndjson") if l.strip()]
print("spans:", len(rows))
for kind, n in collections.Counter(r["kind"] for r in rows).most_common():
    print(" %4d %s" % (n, kind))
for r in rows:
    if r["kind"] == "chat":
        print(" chat max_tokens=%s in=%s out=%s" % (

```



```

        r["request_max_tokens"], r["input_tokens"], r["output_tokens"]))
EOF
# -> spans: 100 (other 91, phase 3, chat 2, tool 2, root 1, agent 1)
# chat max_tokens=1 in=None out=None <- the probe (rejected by a reasoning model)
# chat max_tokens=None in=320 out=471 <- the real call
# ---- 8. tear down ----
curl -s $CURL_OPTS -X DELETE "$SVC/benchmarks/$BENCH/deploy?$SCOPE" \
-H "Authorization: Bearer $TOKEN" -o /dev/null -w 'teardown -> %{http_code}\n' # 204

```

## 6.1 The same thing in one command ( autobench-cli )

`autobench-cli` is a stdlib-only client that performs §6.0 end to end — token, pre-clean, deploy, the readiness-stability and agent-card gates, run, 424 retry, poll, artifact listing and local mirror — and exits non-zero if the run did not succeed.

```

export BM_BASE="https://autobench-rossoctl-system.apps.ykt3.hcp.res.ibm.com"
export BM_ISS="https://keycloak-keycloak.apps.ykt2.hcp.res.ibm.com/realms/rossoctl"
export BM_PASSWORD_FILE="$HOME/.rossoctl-ykt3/benchmark.pass" # chmod 600
export BM_INSECURE=1
export BM_CARD_TEMPLATE="https://{service}-{namespace}.apps.ykt2.hcp.res.ibm.com/.well-known/agent-card.json"
# See "What you need on the client side" above for install options; the shortest is:
# uv pip install "rossoctl-autobench @ git+https://github.com/rossoctl/autobench" --no-deps
autobench-cli all --benchmark gsm8k --tasks 1 --timeout 120 --mirror /tmp/autobench

```

Which prints, for the run above:

```

[17:56:12] agent card 200 (https://exgentic-a2a-tool-calling-gsm8k-team1.apps.ykt2...)
[17:56:28] run_id=20260902215628-79e0713a
[17:56:38] terminal=succeeded summary={"total": 1, ..., "pass_rate": 1.0, "wall_seconds": 5.4}
[17:56:38] artifacts_prefix=ykt3-to-ykt2/benchmark/keycloak-...-rossoctl/gsm8k/20260902215628-79e0713a
      json          468 https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/.../run.json
      ...
[17:56:39] mirrored 8/8 -> /tmp/autobench/ykt3-to-ykt2/.../20260902215628-79e0713a
-rw-r--r--      3002 Sep 02 17:56 manifest.json
-rw-r--r--      1035 Sep 02 17:56 report.ndjson
-rw-r--r--     11984 Sep 02 17:56 report.parquet
-rw-r--r--       468 Sep 02 17:56 run.json
-rw-r--r--     52578 Sep 02 17:56 span_report.ndjson
-rw-r--r--     10878 Sep 02 17:56 span_report.parquet
-rw-r--r--       371 Sep 02 17:56 token_report.ndjson
-rw-r--r--     4425 Sep 02 17:56 token_report.parquet
8 files, 84,741 bytes

cd /tmp/autobench/ykt3-to-ykt2/.../20260902215628-79e0713a

status      succeeded
pass_rate   1.0    (1/1)
wall        5s

```

The trailing `cd` is there because the mirror path is absolute and deeply nested — retyping it relative to your current directory is the easy mistake. The sizes are also the quickest check that nothing arrived truncated. Feed that directory straight into §6.0 step 7 to analyse the run.

Each step is also a subcommand, so the CLI doubles as a way to see one HTTP call at a time:

```

autobench-cli whoami                    # GET /hello
autobench-cli list                      # GET /benchmarks
autobench-cli deploy --benchmark gsm8k
autobench-cli wait --benchmark gsm8k    # stability + card gates
autobench-cli run --benchmark gsm8k --tasks 1 # prints the run_id
autobench-cli poll --benchmark gsm8k --run "$RUN"
autobench-cli report --benchmark gsm8k --run "$RUN" # GET .../report (needs MLflow)
autobench-cli artifacts --benchmark gsm8k --run "$RUN" --mirror /tmp/autobench # + ls -l
autobench-cli teardown --benchmark gsm8k

```

No install? It runs straight from a checkout too — `python -m autobench.cli all ...` (the module imports nothing beyond the standard library, so the server's dependencies are not needed to drive the API).

Every option is catalogued in §6.2. Switching to KinD needs only three lines:

```

export BM_BASE="http://autobench.localtest.me:8080"
export BM_ISS="http://keycloak.localtest.me:8080/realms/rossoctl"
export BM_PASSWORD_FILE="$HOME/.rossoctl-kind/benchmarkers.pass"; unset BM_INSECURE
BM_CARD_TEMPLATE

```

For the full 12-run matrix use `reference/run-12.py` instead — see §6.4. It implements the same gates independently; consolidating both behind `autobench.cli` is a known follow-up. If you want the 12 legs as 12 individual commands in the shape of the line above, they are tabulated in §6.5.

## 6.2 autobench-cli options

`autobench-cli <command> [options]` . The **command** is **positional** and required; everything else is a flag. Defaults below are the real argparse defaults, so a bare `autobench-cli all` runs one gsm8k task in `team1` with a 300s budget.

**Commands.** `all` runs the whole lifecycle; the rest are the individual steps, so the same tool serves "just run it" and "show me one HTTP call".

| Command                        | Does                                                                               |
|--------------------------------|------------------------------------------------------------------------------------|
| <code>all</code>               | pre-clean → deploy → wait → run → poll → artifacts → summary (→ teardown if asked) |
| <code>whoami</code>            | GET /hello — proves the token and shows which instance you routed to               |
| <code>list</code>              | GET /benchmarks                                                                    |
| <code>deploy / teardown</code> | create / delete the MCP tool + A2A agent                                           |
| <code>wait</code>              | block on the readiness-stability and agent-card gates                              |
| <code>run</code>               | POST .../runs, prints the run_id                                                   |
| <code>poll</code>              | follow one run to a terminal status                                                |
| <code>report</code>            | GET .../report (needs MLflow configured, else empty)                               |
| <code>artifacts</code>         | list the run's S3 objects, and mirror them with <code>--mirror</code>              |

### What to run, and how much of it.

| Option                   | Default            | Meaning                              |
|--------------------------|--------------------|--------------------------------------|
| <code>--benchmark</code> | <code>gsm8k</code> | <code>gsm8k   tau2   appworld</code> |

| Option                        | Default      | Meaning                                                                                            |
|-------------------------------|--------------|----------------------------------------------------------------------------------------------------|
| <code>--tasks N</code>        | 1            | <code>max_tasks</code> — <b>how many</b> benchmark problems to attempt (capped by the pool, §5.3)  |
| <code>--parallel N</code>     | 1            | <code>max_parallel_sessions</code> — <b>how many at once</b>                                       |
| <code>--timeout S</code>      | 300          | <code>timeout_seconds</code> — wall budget for the <b>whole run</b>                                |
| <code>--task-timeout S</code> | <i>unset</i> | <code>task_timeout_seconds</code> — ceiling for <b>one</b> task, clamped to <code>--timeout</code> |

`--tasks` and `--parallel` are independent: `--tasks 50 --parallel 4` is fifty problems, four concurrently. Task selection is **deterministic**, so `--tasks 1` runs the *same* problem every time — which is why it is the standard smoke test (a 1-task gsm8k run reproduces 320 input / 87 output tokens). Set `--task-timeout` on multi-turn work: without it one wedged task can consume the entire `--timeout`, which is why the canonical matrix gives tau2 600s per task under a 2100s wall. §5.3 has the semantics in full — what a multi-task run is, and why both over- and under-asking on `--tasks` are silent.

**Where it goes.** `--namespace` (`team1`), `--agent` (`tool_calling`), `--experiment` (`default`) form the scope that `deploy / wait / teardown` address.

**Deploy-time only** — these bake into the workload, so they need a (re)deploy to take effect; a run cannot re-point them.

| Option                            | Meaning                                                                                                     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>--model</code>              | override the agent's LLM, e.g. <code>openai/Azure/gpt-4.1</code>                                            |
| <code>--preset</code>             | <code>AuthBridge plugin_preset</code> : <code>auth-only</code>   <code>ibac-only</code>   <code>full</code> |
| <code>--plugin NAME:POLICY</code> | repeatable per-plugin override, e.g. <code>--plugin ibac:observe</code>                                     |
| <code>--on-error</code>           | <code>chain-default policy</code> : <code>enforce</code>   <code>observe</code>   <code>off</code>          |

**all modifiers.**

| Option                   | Effect                                                                |
|--------------------------|-----------------------------------------------------------------------|
| <code>--no-deploy</code> | skip the pre-clean <b>and</b> the deploy; reuse what is already there |
| <code>--teardown</code>  | delete the deployment when finished                                   |

Note the asymmetry: **all pre-cleans by default** (an unconditional `DELETE` before deploying, so it is repeatable) but **does not tear down** — the deployment is left warm unless you ask.

**Readiness gates.** These exist because Service-reported readiness is not sufficient: on OpenShift the Route 502s for ~10s after the Service says Ready, and an agent whose MCP is not yet serving flaps Ready → CrashLoopBackOff → Ready.

| Option                         | Default                                       | Meaning                                              |
|--------------------------------|-----------------------------------------------|------------------------------------------------------|
| <code>--stable N</code>        | 4                                             | consecutive ready polls required before proceeding   |
| <code>--settle S</code>        | 15, or 45 with <code>--preset/--plugin</code> | pause after ready; longer when a sidecar is injected |
| <code>--poll-interval S</code> | 10                                            | gap between status polls                             |
| <code>--wait-timeout S</code>  | 1800                                          | give up waiting for readiness                        |

**Results.** `--run RUN_ID` selects the run for `poll / report / artifacts` . `--mirror DIR` downloads every artifact under `DIR` and lists them — the artifact URLs are public, so no AWS credentials are involved.

**Connection and auth.** Each of these takes a flag or an environment variable; the env form is usually easier because the same values drive `reference/run-12.py` .

| Flag                    | Env                        | Default                                                                                                                                          |
|-------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--base</code>     | <code>BM_BASE</code>       | <i>(required)</i> Service base URL                                                                                                               |
| <code>--iss</code>      | <code>BM_ISS</code>        | <i>(required)</i> the instance's <b>own</b> issuer — for a cross-cluster setup this is the <b>workload</b> cluster's Keycloak, not the Service's |
| <code>--user</code>     | <code>BM_USER</code>       | benchmarker                                                                                                                                      |
| <code>--client</code>   | <code>BM_CLIENT</code>     | rossocctl                                                                                                                                        |
| <code>--insecure</code> | <code>BM_INSECURE=1</code> | off — set it for OpenShift's self-signed edge routes                                                                                             |

Two settings are **environment-only**, with no flag:

- `BM_PASSWORD_FILE` (a `chmod 600` file) or `BM_PASSWORD` . One is mandatory; the CLI exits rather than prompting. Prefer the file so the secret never reaches your shell history.
- `BM_CARD_TEMPLATE` — when set, `wait` additionally polls the agent card until it returns 200. Needed for cross-cluster runs where the agent is reachable only via an edge Route.

**Exit codes.** `0` success · `7` the run reached a terminal state that was **not** succeeded · `6` a teardown that returned neither 204 nor 404 · `1` a usage or configuration error. A run that legitimately scores `pass_rate 0.0` still exits `0` if its status is `succeeded` — appworld does this by design, so do not treat exit 0 as "the agent solved it".

## 6.3 Other validated flows

These map to the canonical parameterized runs and were validated on `ykt3` with S3 export.

```
# Run #2 – gsm8k, 10 tasks (validated: 9/10 pass)
curl -s -X POST "$SVC/benchmarks/gsm8k/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 10, "max_parallel_sessions": 1, "timeout_seconds": 600}'
# Run #3 – gsm8k, 50 tasks, 4-way concurrency (validated: 42/50, ~190s)
curl -s -X POST "$SVC/benchmarks/gsm8k/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 50, "max_parallel_sessions": 4, "timeout_seconds": 900}'
# Run #9 – tau2, 10 tasks (multi-turn; deploy tau2 first)
curl -s -X POST "$SVC/benchmarks/tau2/deploy" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" -d '{"namespace": "team1"}'
curl -s -X POST "$SVC/benchmarks/tau2/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 10, "max_parallel_sessions": 1, "timeout_seconds": 1200}'
# Run #10 – tau2, 20 tasks, 4-way concurrency (validated: 17/20, ~915s – needs raised timeout)
curl -s -X POST "$SVC/benchmarks/tau2/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 20, "max_parallel_sessions": 4, "timeout_seconds": 1800}'
```

## 6.4 The whole 12-run matrix in one command ( `reference/run-12.py` )

Same idea as §6.1, one level up: instead of a single benchmark run, this drives all **12 canonical parameterized runs** — gsm8k at three sizes, a model swap, the four AuthBridge presets, tau2 at two sizes, appworld at two — deploying each leg fresh, mirroring every artifact, and writing one state file the report generators read.

Unlike `autobench-cli`, this one **is** repo-only: it is a matrix driver tied to `run12_specs.json`, not something you install. Clone first. (For the same 12 legs written out as 12 separate one-line commands, see §6.5.)

```
git clone https://github.com/rossocctl/autobench && cd autobench
# Same five variables as §6.1 — plus a label, which names the state file and the reports.
export BM_BASE=https://autobench-rossocctl-system.apps.ykt3.hcp.res.ibm.com
export BM_ISS=https://keycloak-keycloak.apps.ykt2.hcp.res.ibm.com/realms/rossocctl
export BM_PASSWORD_FILE="$HOME/.rossocctl-ykt3/benchmarkers.pass"
export BM_INSECURE=1
export BM_CARD_TEMPLATE="https://{service}-{namespace}.apps.ykt2.hcp.res.ibm.com/.well-known/agent-card.json"
export BM_LABEL=ocp-dev146
# Space the legs that share prompts past the LLM gateway's completion cache, and interleave the
# cache groups so the gaps overlap real work instead of being slept through. Without these two
# the
# matrix still runs -- and its latency and output-token columns are partly replays. See below.
export BM_CACHE_GAP=900
export BM_ORDER=1,12,2,10,3,11,5,9,6,4,7,8
python3 reference/run-12.py # ~2h10m; log it: ... 2>&1 | tee /tmp/run12-ocp.log
```

**BM\_CACHE\_GAP** defaults to **0**, and that default silently costs you two columns. The gateway caches completions keyed on the request body for **~10 minutes** (measured by survival curve), and task selection is deterministic, so legs #1–#3 and #5–#8 all open with the *same* five gsm8k prompts — a later leg is served the earlier leg's completions, replaying its `usage` and measuring a cache lookup instead of a call. Input tokens and pass rates are unaffected; **per-call latency and output tokens are not comparable across those legs** unless they are spaced. `900` gives a 50% margin, only the *shortfall* is slept, and an interleaving `BM_ORDER` is what keeps the cost down: the v1.28 matrices slept **33 minutes** per platform on top of ~1h35m of leg time, where the spec-file order would have slept about twice that. **Latency is the wrong detector for a replay** — one measured replay took 3.0 s, the same as a miss — so compare response `id`s, not durations.

It ends with a per-leg summary and the state file path:

```
[21:03:57] === done in 7845s; 12 runs -> /tmp/autobench/run12-ocp-dev146.json
[21:03:57] #1 gsm8k succeeded pass=1.0 tasks=1
[21:03:57] #2 gsm8k succeeded pass=1.0 tasks=10
...
[21:03:57] #12 appworld succeeded pass=0.0 tasks=20
```

Then turn that state file into the two documents (all numbers derived from the mirrored artifacts — nothing transcribed):

```
# 1. the 12-run report. <date> is the date the RUNS executed, from the modal run_id prefix.
python3 reference/gen-12run-report.py /tmp/autobench/run12-ocp-dev146.json v1.28 \
  "OpenShift - ykt3 Service, ykt2 workloads (cross-cluster)" \
  results/12run-report-v1.28-20260915-ocp.md
# 2. OCP vs KinD, once both matrices exist
python3 reference/gen-12run-comparison.py \
```

```
/tmp/autobench/run12-ocp-dev146.json "OpenShift (ykt3 Service, ykt2 workloads)" \
/tmp/autobench/run12-kind-dev146.json "KinD (single-node local)" v1.28 \
results/12run-comparison-v1.28-20260915.md
```

**Do not mine plugin overhead out of this matrix.** `reference/gen-plugin-overhead.py` does exactly that — it compares the no-plugin leg #3 against the preset legs #5–#8 — and it is kept only because its source comments record why the approach fails. The canonical matrix runs each preset **once at `max_tasks=5` in a fixed order**, which cannot separate a plugin's cost from run-order drift, and its cross-benchmark projection was falsified **in both directions** (over-estimating by 2.2× on one cluster, under-estimating by 2.2× on the other). Its per-layer figures are withdrawn, not corrected. The designed replacement is a separate 13-leg experiment with its own spec file, reversed-order replicates and `n=50` — see [PLUGIN\\_OVERHEAD.md](#) for the findings and `reference/gen-plugin-study.py` / `gen-plugin-study-xplat.py` for the generators:

```
BM_SPECS=reference/plugin_study_specs.json BM_LABEL=pstudy-ocp-v128 \
BM_CACHE_GAP=900 BM_ORDER=111,112,101,102,103,104,105,113,106,107,108,109,110 \
python3 reference/run-12.py # ~3h40m, most of it cache gaps; detach it
# The judge log is what tells the analyzer how many tool calls were actually authorized –
# without it, per-preset medians mix judged and unjudged calls and mislead.
oc -n rossoctl-system logs deploy/ibac-judge --since=6h --timestamps \
| grep -vi healthz | awk '{print $1}' > /tmp/judge-ocp.ts
python3 reference/gen-plugin-study.py /tmp/autobench/run12-pstudy-ocp-v128.json \
reference/plugin_study_specs.json v1.28 \
"OpenShift – ykt3 Service / ykt2 workloads" \
results/12run-plugin-study-v1.28-20260915-ocp.md /tmp/judge-ocp.ts
```

Switching to KinD is the same three-line change as §6.1 plus a new label:

```
export BM_BASE=http://autobench.localtest.me:8080
export BM_ISS=http://keycloak.localtest.me:8080/realms/rossoctl
export BM_PASSWORD_FILE="$HOME/.rossoctl-kind/benchmarkers.pass"
unset BM_INSECURE BM_CARD_TEMPLATE
export BM_LABEL=kind-dev146
python3 reference/run-12.py # ~2h13m on a single node too
```

**Run the two platforms sequentially, not in parallel.** Each platform has its own litellm deployment — hence its own completion cache and its own key table, which is why a key copied between them always 401s — but the **same upstream model providers** sit behind both. Concurrent runs therefore contend exactly where it matters and the latency numbers stop meaning anything.

Other knobs, all optional:

| Variable     | Default                    | Use                                                                                                                                                                                   |
|--------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BM_CACHE_GAP | 0                          | seconds a (benchmark, model) prompt set must rest between legs so the gateway's completion cache expires. <b>Set it to 900</b> — the default measures replays on legs #1–#3 and #5–#8 |
| BM_ORDER     | the spec file's            | comma-separated execution order. Interleave the cache groups (1, 12, 2, 10, 3, 11, 5, 9, 6, 4, 7, 8) so BM_CACHE_GAP overlaps real work: 33 min slept instead of ~68                  |
| BM_SPECS     | reference/run12_specs.json | an alternate spec file, so an experiment reuses this driver without touching the canonical matrix. Accepts a bare list of legs or {"order": [...], "legs": [...]}                     |



| Variable                             | Default                | Use                                                                                                                                                    |
|--------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| BM_ONLY                              | all 12                 | comma-separated leg numbers, e.g. BM_ONLY=7 to re-run one leg, BM_ONLY=1,2,3 for a smoke pass                                                          |
| BM_STABLE_POLLS                      | 4                      | consecutive ready polls required before a run is submitted                                                                                             |
| BM_SETTLE_PLAIN / BM_SETTLE_SIDE CAR | 15 / 45                | post-ready settle seconds; sidecar deploys need longer                                                                                                 |
| BM_USER / BM_CLIENT                  | benchmarker / rossoctl | non-default realm identity                                                                                                                             |
| BM_PASSWORD                          | —                      | the password inline, instead of BM_PASSWORD_FILE. Prefer the file (chmod 600): an exported variable leaks into env, process listings and shell history |

**BM\_ONLY** is how you repair a matrix without redoing it: a leg that failed to deploy can be re-run on its own and merged into the state file, which is exactly what happened to leg #7 of the v1.24 OCP matrix (a transient `DELETE -> 500` left a stale tool behind and the deploy hit `409`).

## 6.5 The 12 legs as individual `autobench-cli` commands

Each leg of §6.4's matrix is also one `autobench-cli all` line, in the shape of the §6.1 example. Use these to re-run a single leg, to bisect a failure, or to read the matrix as parameters rather than as JSON. Export the same five variables as §6.1 first; every line then needs nothing else.

| #  | One-line command                                                                                                                                                     | What it varies               |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 1  | <code>autobench-cli all --benchmark gsm8k --tasks 1 --timeout 120 --mirror /tmp/autobench</code>                                                                     | baseline — the smoke test    |
| 2  | <code>autobench-cli all --benchmark gsm8k --tasks 10 --timeout 300 --mirror /tmp/autobench</code>                                                                    | volume, still serial         |
| 3  | <code>autobench-cli all --benchmark gsm8k --tasks 50 --parallel 4 --timeout 400 --mirror /tmp/autobench</code>                                                       | volume + concurrency         |
| 4  | <code>autobench-cli all --benchmark gsm8k --model openai/Azure/gpt-4.1 --tasks 5 --parallel 4 --timeout 300 --task-timeout 120 --mirror /tmp/autobench</code>        | model swap                   |
| 5  | <code>autobench-cli all --benchmark gsm8k --preset auth-only --tasks 5 --parallel 4 --timeout 300 --task-timeout 120 --mirror /tmp/autobench</code>                  | AuthBridge: auth-only        |
| 6  | <code>autobench-cli all --benchmark gsm8k --preset ibac-only --tasks 5 --parallel 4 --timeout 300 --task-timeout 120 --mirror /tmp/autobench</code>                  | AuthBridge: ibac-only        |
| 7  | <code>autobench-cli all --benchmark gsm8k --preset full --tasks 5 --parallel 4 --timeout 300 --task-timeout 120 --mirror /tmp/autobench</code>                       | AuthBridge: full (enforce)   |
| 8  | <code>autobench-cli all --benchmark gsm8k --preset full --plugin ibac:observe --tasks 5 --parallel 4 --timeout 300 --task-timeout 120 --mirror /tmp/autobench</code> | full + per-plugin override   |
| 9  | <code>autobench-cli all --benchmark tau2 --tasks 10 --timeout 2100 --task-timeout 600 --mirror /tmp/autobench</code>                                                 | multi-turn + user simulator  |
| 10 | <code>autobench-cli all --benchmark tau2 --tasks 20 --parallel 4 --timeout 2400 --task-timeout 600 --mirror /tmp/autobench</code>                                    | multi-turn under concurrency |



| #  | One-line command                                                                                                                                                    | What it varies                 |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 11 | <code>autobench-cli all --benchmark appworld --model openai/gemini-2.5-pro --tasks 5 --timeout 3300 --task-timeout 600 --mirror /tmp/autobench</code>               | long-horizon                   |
| 12 | <code>autobench-cli all --benchmark appworld --model openai/gemini-2.5-pro --tasks 20 --parallel 4 --timeout 3600 --task-timeout 600 --mirror /tmp/autobench</code> | long-horizon under concurrency |

These are generated from `reference/run12_specs.json`, not transcribed: each line's run body round-trips to the spec's `run` object field for field, and `--model` / `--preset` / `--plugin` reproduce the spec's deploy body as well (`--preset` implies `authbridge_enabled: true`; the CLI also sends `experiment: "default"`, which is the server's default anyway). Leg #1 is the §6.1 example unchanged. Two details the table encodes silently:

- **Every line is a fresh deploy.** `all` pre-cleans with `DELETE .../deploy` before deploying, which is what the matrix requires. That is also why `--no-deploy` has no place here. The reason has changed as of `exgentic 0.3.5.dev145` but the rule has not. Warm reuse no longer drops the usage-bearing span — that defect is fixed, measured. What a fresh deploy still buys is a **fresh image pull**: these workloads are pinned to `:latest` under `imagePullPolicy: Always`, so a newly created pod picks up an upstream fix while a long-lived one serves a six-week-old digest indefinitely. It does *not* protect you from the LLM gateway's own response cache, which is outside the agent and unavoidable from here. See `docs/exgentic-agent-bug-report-20260901.md`.
- **--settle needs no value.** It defaults to 45 s when `--preset` or `--plugin` is present and 15 s otherwise, matching `BM_SETTLE_SIDE CAR` / `BM_SETTLE_PLAIN` in §6.4.

**Run them one at a time, in order.** Legs sharing a benchmark share one deployment slot, so two of these in parallel will fight over it; and legs #3 and #5–#8 are only comparable because they execute byte-identical work, which a concurrent gateway load would spoil. Each exits `0` on `succeeded` and `7` otherwise (§6.2), so `&&` chaining stops at the first bad leg.

These do **not** replace `reference/run-12.py`. The driver writes `/tmp/autobench/run12-<BM_LABEL>.json`, and that state file is the only input the three report generators accept — run the legs by hand and you get the artifacts but none of the documents. Use §6.4 for a real matrix; use this table to re-run or inspect one leg of it.

## 7. Known limits & error codes

| Situation                                                                                | Code | What to do                                                                                          |
|------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------|
| Missing/invalid bearer                                                                   | 401  | fetch a fresh token (§2)                                                                            |
| <code>iss</code> not in any instance file                                                | 403  | add/repair the instance config (§3.1)                                                               |
| <code>/config</code> as non-benchmarkers                                                 | 403  | authenticate as the benchmarkers user                                                               |
| Setting a workload cred via <code>/config</code>                                         | 422  | provision it as a cluster Secret instead (§3.3)                                                     |
| <code>plugin_preset/plugins/on_error</code> without <code>authbridge_enabled=true</code> | 422  | set <code>authbridge_enabled=true</code> so the sidecar is injected for the pipeline to take effect |

| Situation                        | Code | What to do                                                                       |
|----------------------------------|------|----------------------------------------------------------------------------------|
| plugin_config_file on deploy     | 422  | local-path input with no HTTP analog; use plugin_preset/plugins/on_error instead |
| Run before deploy                | 409  | POST .../deploy first                                                            |
| Run while not Ready              | 424  | provision the named Secret(s), wait for Ready (§5.2)                             |
| Report with MLflow unconfigured  | 409  | set MLflow read creds via PUT /config (§3.4)                                     |
| Upstream Rossoctl/MLflow failure | 502  | transient upstream issue; retry                                                  |

**AuthBridge plugin presets (runs 4–8 of the ibac comparison):** layer-3 plugin composition and `on_error` selection require overlaying the per-agent `authbridge-config-<agent>` ConfigMap via `kubect`l — which the HTTP-only Service cannot do — so those are honestly rejected with `422`. The one enactable knob is `authbridge_enabled=true` (injects the sidecar with the cluster-default pipeline).

**appworld:** registers and deploys correctly via the Service, but its MCP image hangs on its own per-task Venv-service health timeout (120s), independent of the Service. Treat appworld as externally blocked until that image is fixed.

## 8. Extending the catalog (adding or changing a benchmark)

The benchmark catalog is **code, not runtime data** — a static `BENCHMARKS` dict in `src/autobench/benchmarks/registry.py`. There is no database, config file, or admin API behind it, and the HTTP surface over the catalog is **read-only** (`GET /benchmarks`, `GET /benchmarks/{name}`). Adding or changing a benchmark is therefore a **source change + tests + image rebuild + redeploy**, deliberately: the definition pins container images, dataset env, and resource limits that must move in lockstep with the workload images, so every change is a reviewable, git-tracked, image-versioned artifact rather than mutable state that could drift per instance.

See the co-located contributor note `src/autobench/benchmarks/README.md` for the field-by-field reference and the per-benchmark env gotchas.

### 8.1 What a definition holds

Each entry is a `BenchmarkDefinition`: `name`, `mcp_image` (+ `mcp_image_tag` / `mcp_port` / `mcp_path`), `tool_env`, `tool_resources`, `default_model`, `user_simulator` (multi-turn flag), and an `agents` map of `BenchmarkAgentSpec` (per-agent `container_image` + `extra_env` + `resources`). Secret references are declared with the `_secret_env(env, secret, key)` helper; `required_secrets()` derives the actionable `424` precheck message from them automatically.

### 8.2 Add a new benchmark

1. Add a `BenchmarkDefinition` entry to `BENCHMARKS` in `registry.py`:
  - `mcp_image` (+ tag/port/path) — the MCP tool image.
  - `tool_env` — `BENCHMARK_NAME` and any secret refs via `_secret_env`. The LLM base is **not** baked in: it comes from the instance's required `workload_llm.api_base` (KinD must use ETE's internal `...vpc-int...` endpoint; a deploy with no gateway configured is rejected with `422`).

- `agents={...}` — one `BenchmarkAgentSpec` per flavor. The `tool_calling` A2A image is shared across all current benchmarks; usually you reuse it and only the name gets a `-<benchmark>` suffix.
- `user_simulator=True` **only** for multi-turn benchmarks — this is what makes `build_tool_request` inject `EXGENTIC_SET_BENCHMARK_USER_SIMULATOR_MODEL` into the MCP pod so the server-side simulator shares the run's model.
- Any benchmark-quirk env — mind the documented gotchas: gsm8k needs `EXGENTIC_SET_BENCHMARK_RUNNER=direct`; tau2 needs `EXGENTIC_SET_BENCHMARK_ACTION_TIMEOUT=1000`; appworld **rejects** that same action-timeout override and crashes at startup if it's present.

2. **Add tests** in `tests/test_benchmarks.py` — assert `build_tool_request` / `build_agent_request` emit the expected images, env, secrets, and resources, mirroring the existing patterns.
3. **Rebuild + bump the image tag**, redeploy the Service. The new benchmark then appears in `GET /benchmarks` and is deployable/runnable at `/benchmarks/<newname>/...` with **no client change** — the request bodies are benchmark-agnostic (see §5).

### 8.3 Change an existing benchmark

Same mechanism — edit the entry (bump `mcp_image_tag`, change `default_model`, add an env var, adjust `tool_resources`), update tests, rebuild, redeploy. Because `required_secrets()` is derived from `tool_env` + the chosen agent's `extra_env`, changing a secret reference automatically updates the 424 "this benchmark requires secret(s): ..." precheck message — no separate wiring.

### 8.4 What stays runtime-mutable (for contrast)

Per-instance state is *not* in the catalog and does not require a rebuild: instance config (files under `settings.instances_dir`, keyed by `iss`) and the benchmarker-only `PUT /config` overrides (MLflow read creds + S3). The benchmark catalog is intentionally the immutable, version-pinned part.

## 9. Regenerating the documents

Everything under `docs/` that is not hand-written has a generator, so no step in producing the published artifacts is a manual browser print or a hand-edited table.

| Artifact                                                                               | Generated by                                                                      | Source of truth                                                |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------|
| <code>docs/DEVELOPER_GUIDE.pdf</code> ,<br><code>docs/12_RUNS_CROSS_CLUSTER.pdf</code> | <code>python3 reference/gen_pdf.py</code>                                         | the <code>.md</code> beside it                                 |
| the table of contents in this file                                                     | <code>python3 reference/gen_toc.py</code><br><code>docs/DEVELOPER_GUIDE.md</code> | this file's own headings                                       |
| §6.5's 12 one-line commands                                                            | <code>reference/run12_specs.json</code>                                           | the same specs<br><code>reference/run-12.py</code><br>executes |
| <code>docs/AutoBench.pptx</code>                                                       | <code>uv run --with python-pptx python docs/generate_pptx.py</code>               | <code>results/12run-*.md</code> + this guide                   |

| Artifact           | Generated by                                  | Source of truth     |
|--------------------|-----------------------------------------------|---------------------|
| docs/AutoBench.pdf | python3 reference/gen_pdf.py<br>(LibreOffice) | docs/AutoBench.pptx |

`gen_pdf.py` with no arguments rebuilds all three PDFs (~10 s). It dispatches on extension: `.md` goes through `pandoc` → CSS → headless Chrome, `.pptx` through LibreOffice. Pass paths to do a subset, `-o` to write elsewhere:

```
python3 reference/gen_pdf.py                # all three
python3 reference/gen_pdf.py docs/DEVELOPER_GUIDE.md    # just this guide
python3 reference/gen_pdf.py docs/12_RUNS_CROSS_CLUSTER.md -o /tmp/draft.pdf
```

**Every generated PDF carries a bookmarks outline** (the navigation sidebar), and each run prints what it produced — `outline: 36 bookmarks, depth 0-3, first='...'` — so a silent regression to a PDF you cannot navigate shows up in the build output instead of in a reader's lap. Two different mechanisms produce it, which is why both needed fixing:

- `.md` → **Chrome** builds the outline from the `h1 – h6` tree, but only with `--generate-pdf-document-outline` (Chrome ≥ 122). Without that switch Chrome emits no outline at all, and it ignores the unknown flag silently on older builds.
- `.pptx` → **LibreOffice** names each bookmark after the slide's **name**, not its title text. `python-pptx` leaves `<p:cSld name>` unset, so the deck's outline read `Slide 1 ... Slide 16`; `generate_pptx.py` now names every slide after its own title in a final pass, next to the page-numbering pass. Regenerate the `.pptx` before the `.pdf` or the old names persist.

The CSS lives in the script rather than in a theme file for one reason worth knowing before you change it: **long code lines must wrap, not clip**. A 190-char `autobench-cli` line or a 185-char S3 URL runs off the page under default print styling, and the flags that matter are at the end. Three rules prevent that — `white-space: pre-wrap`, a hanging indent per *logical* source line (so a wrapped continuation is visibly a continuation and not a second command), and `overflow-wrap` where there are no spaces to break at. Long table cells opt into breaking mid-word by measured length; applying it to every cell shrinks narrow columns until two-digit numbers stack vertically.

**Every artifact in `docs/` is covered by the table above, and it is worth keeping that way: if no generator rebuilds it, don't commit it.** A hand-made export does not merely go out of date — it keeps publishing what the current text has retracted, and nothing in the build output will tell you. That is why there is no Word copy of this guide and no rendered PDF of the OpenAPI documents. If you need another format, convert on demand (`pandoc docs/DEVELOPER_GUIDE.md -o /tmp/guide.docx`) and leave the result out of the repo. The OpenAPI documents themselves — `docs/openapi.json` / `docs/openapi.yaml` — are dumps of the schema `_install_openapi()` builds in `src/autobench/app.py` rather than hand-written text, but no dump script is committed either: refresh them from a running Service's `/openapi.json`.